Sankaraleengam (Sankar) Alagapan

School of Electrical and Computer Engineering
Georgia Institute of Technology
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Research Interests

Human Neuroscience Brain Stimulation, NeuroAI

Education

Ph.D. Biomedical Engineering, University of Florida, Gainesville, FL — 2014
Dissertation: Information propagation in engineered neuronal networks

M.S. Biomedical Engineering, University of Florida, Gainesville, FL — 2008

B.E. Electrical and Electronics Engineering, Anna University, India — 2007

Professional Appointments

Senior Research Scientist, Georgia Institute of Technology — 2026 – Present
Research Scientist II, Georgia Institute of Technology — 2022 – 2026
Research Engineer II, Georgia Institute of Technology — 2018 – 2022
Postdoctoral Research Associate, UNC Chapel Hill — 2014 – 2018
Graduate Research Assistant, University of Florida — 2008 – 2014

Honors & Awards

Georgia Institute of Technology
Georgia CTSA KL2 Scholar — 2023
Sigma Xi Best Faculty Paper Award — 2024
Roger P. Webb Outstanding Non-Tenure Faculty — 2024

University of North Carolina at Chapel Hill
Helen Lyng White Postdoctoral Fellowship — 2016
Office of Postdoctoral Affairs Travel Award — 2016

Research Support

A naturalistic multimodal platform for capturing brain-body interactions during physical effort-based decision making Role: Co-Investigator	NIH NIMH	2025–2028
Multimodal Biometrics to Optimize and Scale Decision Support for Depression DBS Role: Co-Investigator	NIH NINDS	2025–2030
Georgia Clinical and Translational Science Alliance Role: KL2 Scholar	NIH NCATS	2023–2025

Mentoring

Postdoctoral Fellows

Parisa Sarikhani	*Biophysical models to inform clinical decision making in DBS for depression* Georgia CTSA TL1 and Presidential Postdoctoral Fellow Co-advised with Chris Rozell	2024–Present
Ermyntrude Adjei	*Multimodal interactions in neuropsychiatric disorders* Co-advised with Chris Rozell	2026–Present

PhD Students

Shreesh Karjagi	*Developing objective markers for biometric decision Support for DBS for depression* Co-advised with Chris Rozell	2025–Present
Vivek Anand	*Measuring subjective effort using adaptive sampling* NDSEG Fellow CNTP Training Fellow Co-advised with Chris Rozell	2023–Present
Tyler Albarran	*Brain-body interactions during naturalistic effort-based decision-making* CNTP Training Fellow Co-advised with Chris Rozell	2023–Present
Elif Ceren Fitoz	*Dissertation: Brain-based biomarkers for SCC DBS: Longitudinal and evoked neural signatures of depression recovery* Co-advised with Chris Rozell	2022–Present

Master's Students

Dong Yeun Kim	*Capstone: Interpretable machine learning for potential biomarkers for epilepsy*	2025–2026
Vikram Kaushik	*Decomposed linear dynamical systems for multimodal data*	2024–2025

Research Staff

Camilla May	Research Technician	2023–2026
Emir Akat	Research Technician	2026–Present

Dissertation Committees

Aryaman Gala	Ph.D. Neuroscience (Emory)	2026–
Xuanzhou Chen	Ph.D. ML (Georgia Tech	2026–

Publications

Preprints

1. Anand, V., Helbling, A., Davenport, M. A., Berman, G. J., Alagapan, S. & Rozell, C. J. *LORE: Jointly Learning the Intrinsic Dimensionality and Relative Similarity Structure From Ordinal Data* arXiv:2602.04192 [cs.LG]. http://arxiv.org/abs/2602.04192 (2026) (Feb. 2026).
2. Fitoz, E. C., Alagapan, S., Cha, J., Choi, K. S., Figee, M., Kopell, B., Obatusin, M., Heisig, S., Nauvel, T. & Razavilar, A. *Common Electrophysiology Biomarkers Collected at Home Robustly Track Depression Recovery With Deep Brain Stimulation* https://www.medrxiv.org/content/10.64898/2026.04.13.26350107.abstract (2026) (2026).
3. Desai, A., Van Bogaert, T., Choi, K. S., Song, H. N., Cha, J., Alagapan, S., Xu, E., Dahill-Fuchel, J., Veerakumar, A. & Kopell, B. H. *Longitudinal Changes in Cortical Response Dynamics with Deep Brain Stimulation to the Subcallosal Cingulate for Treatment-Resistant Depression* https://www.biorxiv.org/content/10.1101/2025.11.26.690543.abstract (2026) (2025).

Journal Articles

4. Pinotsis, D. A., Alagapan, S., Sarikhani, P., Nauvel, T., Rozell, C. J. & Mayberg, H. S. Ephaptic coupling and power fluctuations in depression. *Cerebral Cortex* **36,** bhag019. ISSN: 1460-2199. https://doi.org/10.1093/cercor/bhag019 (2026) (Mar. 2026).
5. Xu, E., Pitts, S., Dahill-Fuchel, J., Scherrer, S., Nauvel, T., Overton, J. G., Riva-Posse, P., Crowell, A., Figee, M., Alagapan, S., *et al.* Neural Interoceptive Processing Is Modulated by Deep Brain Stimulation to Subcallosal Cingulate Cortex for Treatment-Resistant Depression. *Biological Psychiatry: Cognitive Neuroscience and Neuroimaging* **10,** 495–503 (2025).
6. Van Rheede*, J. J., Alagapan*, S., Denison, T. J., Riva-Posse, P., Rozell, C. J., Mayberg, H. S., Waters, A. C. & Sharott, A. Cortical signatures of sleep are altered following effective deep brain stimulation for depression. *Translational Psychiatry* **14,** 103 (2024).
7. Alagapan, S., Choi, K. S., Heisig, S., Riva-Posse, P., Crowell, A., Tiruvadi, V., Obatusin, M., Veerakumar, A., Waters, A. C., Gross, R. E., *et al.* Cingulate dynamics track depression recovery with deep brain stimulation. *Nature* **622,** 130–138 (2023).
8. Smith, E. E., Choi, K. S., Veerakumar, A., Obatusin, M., Howell, B., Smith, A. H., Tiruvadi, V., Crowell, A. L., Riva-Posse, P., Alagapan, S., *et al.* Time-frequency signatures evoked by single-pulse deep brain stimulation to the subcallosal cingulate. *Frontiers in Human Neuroscience* **16,** 939258 (2022).
9. Alagapan, S., Waters, A., Veerakumar, A., Obatusin, M., Tiruvadi, V., Crowell, A., Riva-Posse, P., Butera, R., Mayberg, H. & Rozell, C. Longitudinal changes in subcallosal cingulate local field potential features in patients undergoing DBS for treatment-resistant depression. *Biological Psychiatry* **87,** S193 (2020).
10. Alagapan, S., Riddle, J., Huang, W. A., Hadar, E., Shin, H. W. & Fr□hlich, F. Network-Targeted, Multi-site Direct Cortical Stimulation Enhances Working Memory by Modulating Phase Lag of Low-Frequency Oscillations. *Cell Reports* **29,** 2590–2598 (2019).
11. Alagapan, S., Shin, H. W., Fröhlich, F. & Wu, H.-t. Diffusion geometry approach to efficiently remove electrical stimulation artifacts in intracranial electroencephalography. *Journal of neural engineering* **16,** 036010 (2019).
12. Alexander*, M. L., Alagapan*, S., Lugo, C. E., Mellin, J. M., Lustenberger, C., Rubinow, D. R. & Fröhlich, F. Double-blind, randomized pilot clinical trial targeting alpha oscillations with transcranial alternating current stimulation (tACS) for the treatment of major depressive disorder (MDD). *Translational psychiatry* **9,** 1–12 (2019).
13. McPherson, T., Berger, D., Alagapan, S. & Fröhlich, F. Active and passive rhythmic music therapy interventions differentially modulate sympathetic autonomic nervous system activity. *Journal of music therapy* **56,** 240–264 (2019).
14. Ahn, S., Mellin, J. M., Alagapan, S., Alexander, M. L., Gilmore, J. H., Jarskog, L. F. & Fröhlich, F. Targeting reduced neural oscillations in patients with schizophrenia by transcranial alternating current stimulation. *NeuroImage* **186,** 126–136 (2018).
15. Alagapan, S., Lustenberger, C., Hadar, E., Shin, H. W. & Fr□hlich, F. Low-frequency direct cortical stimulation of left superior frontal gyrus enhances working memory performance. *NeuroImage* **184,** 697–706 (2018).
16. McPherson, T., Berger, D., Alagapan, S. & Fröhlich, F. Intrinsic rhythmicity predicts synchronization continuation entrainment performance. *Scientific Reports* **8,** 11782 (2018).
17. Mellin, J. M., Alagapan, S., Lustenberger, C., Lugo, C. E., Alexander, M. L., Gilmore, J. H., Jarskog, L. F. & Fröhlich, F. Randomized trial of transcranial alternating current stimulation for treatment of auditory hallucinations in schizophrenia. *European Psychiatry* **51,** 25–33 (2018).

18. Sheffield, A., Ahn, S., Alagapan, S. & Fröhlich, F. Modulating neural oscillations by transcranial static magnetic field stimulation of the dorsolateral prefrontal cortex: A crossover, double-blind, sham-controlled pilot study. *European Journal of Neuroscience* **49,** 250–262 (2018).
19. Lustenberger, C., Patel, Y. A., Alagapan, S., Page, J. M., Price, B., Boyle, M. R. & Fröhlich, F. High-density EEG characterization of brain responses to auditory rhythmic stimuli during wakefulness and NREM sleep. *NeuroImage* **169,** 57–68 (2017).
20. Alagapan, S., Franca, E., Pan, L., Leondopulos, S., Wheeler, B. C. & DeMarse, T. B. Structure, function, and propagation of information across living two, four, and eight node degree topologies. *Frontiers in Bioengineering and Biotechnology* **4,** 15 (2016).
21. Alagapan, S., Schmidt, S. L., Lefebvre, J., Hadar, E., Shin, H. W. & Fr□hlich, F. Modulation of cortical oscillations by low-frequency direct cortical stimulation is state-dependent. *PLoS biology* **14,** e1002424 (2016).
22. DeMarse, T. B., Pan, L., Alagapan, S., Brewer, G. J. & Wheeler, B. C. Feed-forward propagation of temporal and rate information between cortical populations during coherent activation in engineered in vitro networks. *Frontiers in neural circuits* **10,** 32 (2016).
23. Franca, E., Jao, P. F., Fang, S.-P., Alagapan, S., Pan, L., Yoon, J. H., Yoon, Y.-K. & Wheeler, B. C. Scale of carbon nanomaterials affects neural outgrowth and adhesion. *IEEE transactions on nanobioscience* **15,** 11–18 (2016).
24. Lustenberger, C., Boyle, M. R., Alagapan, S., Mellin, J. M., Vaughn, B. V. & Fröhlich, F. Feedback-controlled transcranial alternating current stimulation reveals a functional role of sleep spindles in motor memory consolidation. *Current Biology* **26,** 2127–2136 (2016).
25. Sellers, K. K., Yu, C., Zhou, Z. C., Stitt, I., Li, Y., Radtke-Schuller, S., Alagapan, S. & Fröhlich, F. Oscillatory dynamics in the frontoparietal attention network during sustained attention in the ferret. *Cell reports* **16,** 2864–2874 (2016).
26. Pan*, L., Alagapan*, S., Franca, E., Leondopulos, S. S., DeMarse, T. B., Brewer, G. J. & Wheeler, B. C. An in vitro method to manipulate the direction and functional strength between neural populations. *Frontiers in neural circuits* **9,** 32 (2015).
27. Brewer, G. J., Boehler, M. D., Leondopulos, S., Pan, L., Alagapan, S., DeMarse, T. & Wheeler, B. C. Toward a self-wired active reconstruction of the hippocampal trisynaptic loop: DG-CA3. *Frontiers in neural circuits* **7,** 165 (2013).
28. Pan, L., Alagapan, S., Franca, E., DeMarse, T., Brewer, G. J. & Wheeler, B. C. Large extracellular spikes recordable from axons in microtunnels. *IEEE Transactions on Neural Systems and Rehabilitation Engineering* **22,** 453–459 (2013).
29. Pan, L., Alagapan, S., Franca, E., Brewer, G. J. & Wheeler, B. C. Propagation of action potential activity in a predefined microtunnel neural network. *Journal of neural engineering* **8,** 046031 (2011).

* *Contributed equally*

Conference Proceedings

30. Martinez-Nunez, A. E., Rozell, C. J., Little, S., Tan, H., Schmidt, S. L., Grill, W. M., Pajic, M., Turner, D. A., de Hemptinne, C., Machado, A., *et al. Proceedings of the 12th annual deep brain stimulation think tank: cutting edge technology meets novel applications* in *Frontiers in Human Neuroscience* **19** (Frontiers Media SA, 2025), 1544994.
31. Chen, Y., Mudrik, N., Johnsen, K. A., Alagapan, S., Charles, A. S. & Rozell, C. *Probabilistic decomposed linear dynamical systems for robust discovery of latent neural dynamics* in *Advances in Neural Information Processing Systems* **37** (2024), 104443–104470.

32. Fitoz, E. C., Alagapan, S., Waters, A., Tiruvadi, V., Veerakumar, A., Obatusin, M., Choi, K. S., Crowell, A., Riva-Posse, P., Butera, R., *et al. Exploration of acute effects of stimulation frequency on subcallosal cingulate dynamics in SCC DBS* in *2023 11th International IEEE/EMBS Conference on Neural Engineering (NER)* (IEEE, 2023), 1–4.
33. Wong, J. K., Deuschl, G., Wolke, R., Bergman, H., Muthuraman, M., Groppa, S., Sheth, S. A., Bronte-Stewart, H. M., Wilkins, K. B., Petrucci, M. N., *et al. Proceedings of the ninth annual deep brain stimulation think tank: advances in cutting edge technologies, artificial intelligence, neuromodulation, neuroethics, pain, interventional psychiatry, epilepsy, and traumatic brain injury* in *Frontiers in Human Neuroscience* **16** (Frontiers Media SA, 2022), 813387.

Patents

34. Alagapan, S., Rozell, C., Mayberg, H., Tiruvadi, V., Waters, A., Riva-Posse, P., Heisig, S., Butera, R., Choi, K. S., Crowell, A., *et al.* (2024).

Presentations

Talks

BIONIC Adaptive Neuromodulation and BCI Symposium, Rochester, MN	2026
ANT NeuroMeeting, Philadelphia, PA	2026
CNS 2025 Workshop on Explainable AI (Virtual)	2025
Minnesota Neuromodulation Symposium, Minneapolis, MN	2025
6th Bioelectronic Medicine Summit, New York, NY	2025
Neurology Grand Rounds, Emory University, Atlanta, GA	2024
O'Donnell Brain Institute, UT Southwestern, Dallas, TX	2024
Psychology Dept Seminar, Emory University, Atlanta, GA	2023
Society for Neuroscience Annual Meeting (Nanosymposium), Washington DC	2023
Society for Neuroscience Annual Meeting (Nanosymposium), San Diego, CA	2022
Society of Biological Psychiatry Annual Meeting (Symposium), New Orleans, LA	2022
Transcranial Electric Stimulation Workshop (Carolina Neurostimulation Conference), Chapel Hill, NC	2018
Society for Industrial and Applied Mathematics Southeast Atlantic Section Conference, Chapel Hill, NC	2018
Neuroscience Center Seminar Series, UNC Chapel Hill, NC	2017

Service

Review panelist, National Science Foundation	2025
Reviewer, MEXA Accelerator	2025
Nanosymposium chair, SfN Annual Meeting	2022
Conference program committee, Carolina Neurostimulation Conference	2018
Ad hoc reviewer - Nature, Science, Nature Communications, Biological Psychiatry, Biological Psychiatry: Cognitive Neuroscience and Neuroimaging, Journal of Computational Neuroscience, Psychological Medicine, Scientific Re-ports, Psychiatry Research, Neuroscience Letters, Journal of Neurosci-ence Methods, NeuroImage, Frontiers in Neuroscience, IEEE Journal of Biomedical and Health Informatics	2016 –